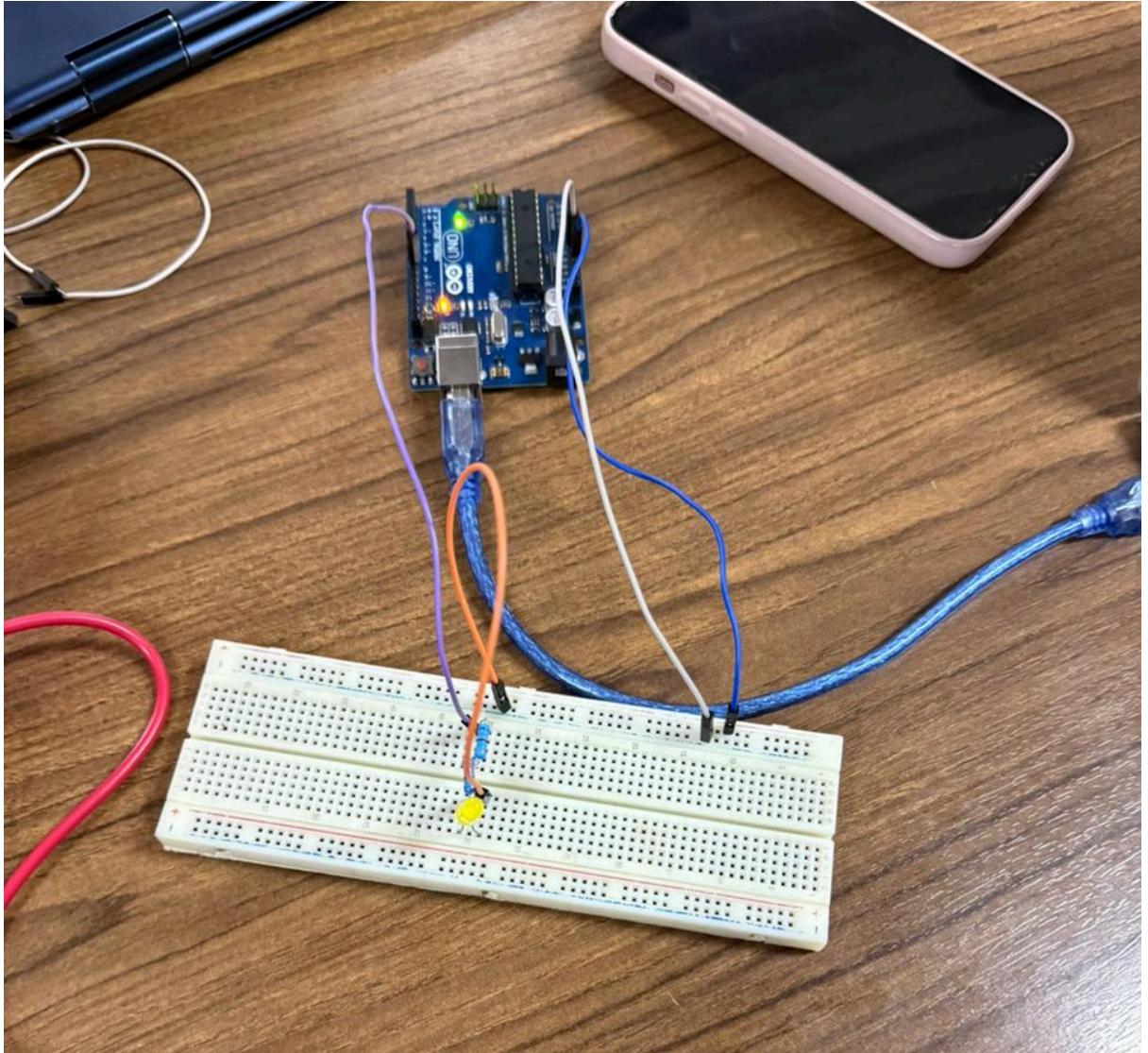


### Experiment 3

#### 1. Physical implementation:

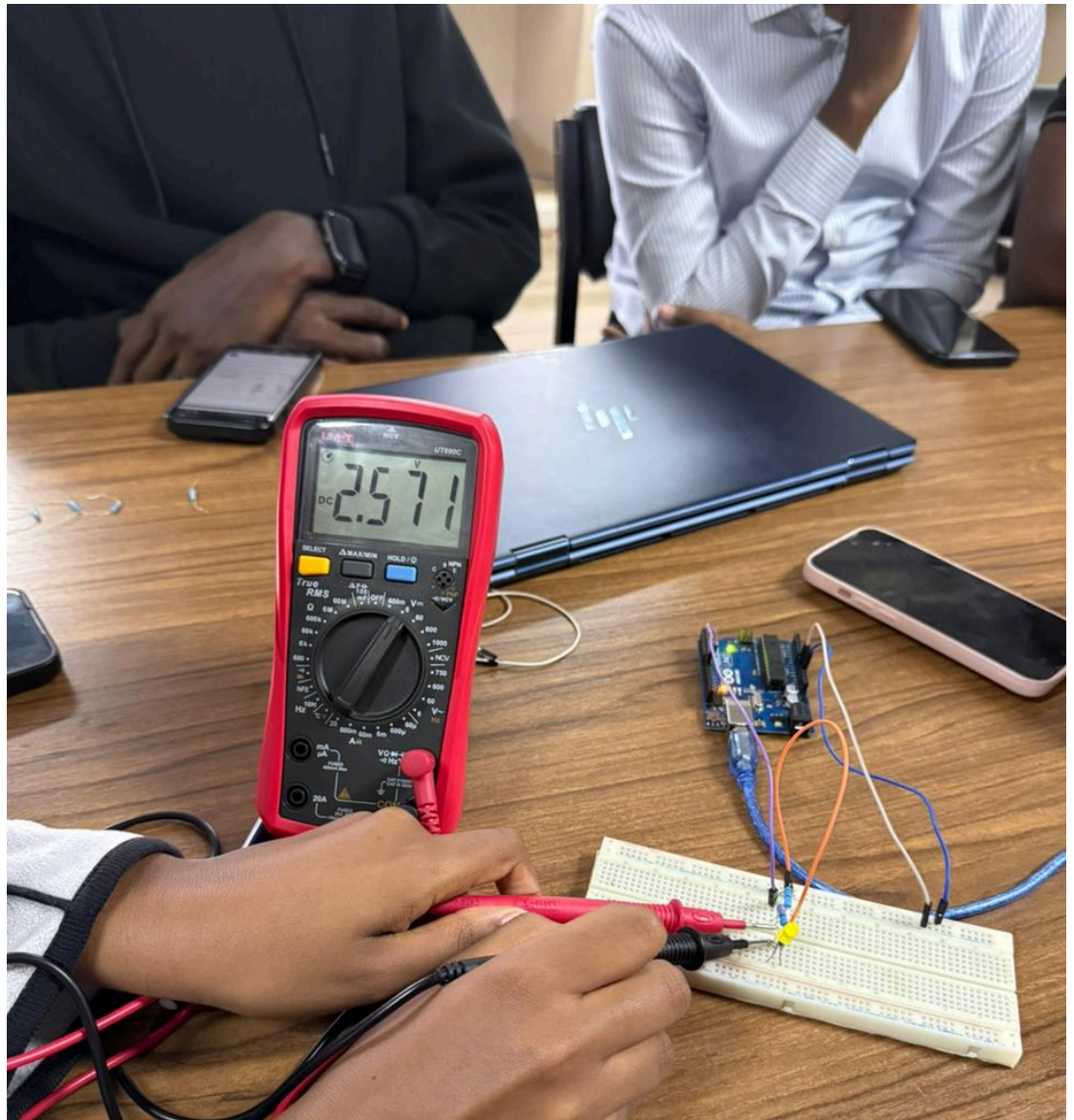


#### 2. Blink program deployed:

```
void setup()
{
  pinMode(LED_BUILTIN, OUTPUT);
}

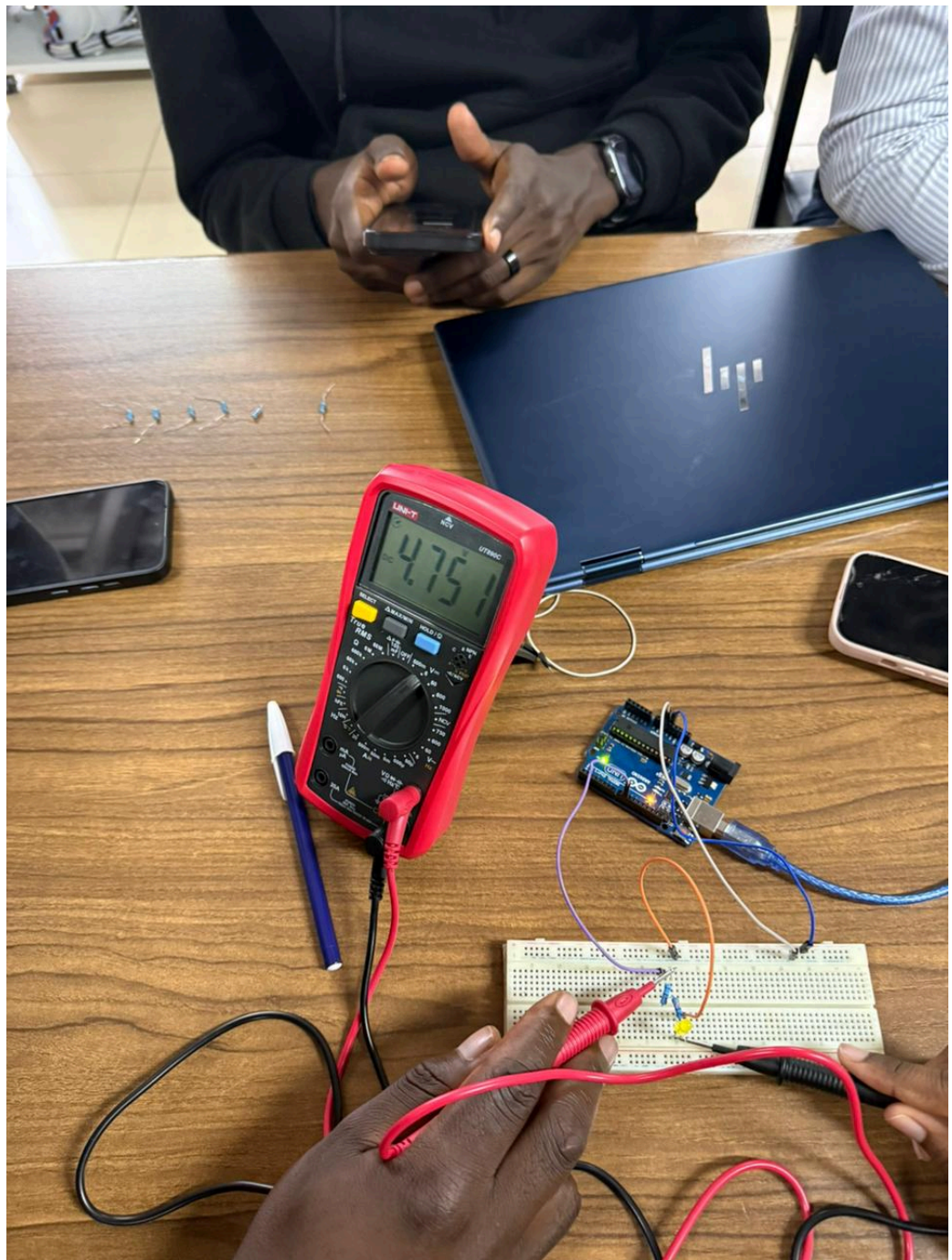
void loop()
{
  digitalWrite(LED_BUILTIN, HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(LED_BUILTIN, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
```

3. Values at resistors:
  - a. First resistor:





b. Second resistor:



4. Calculations using Ohm's Law:

Experiment 3:

i)  $V = IR$

$$\frac{2.571}{330} = \frac{I \times 330}{330}$$

$$I = \underline{\underline{0.0078A}}$$

ii)  $V = IR$

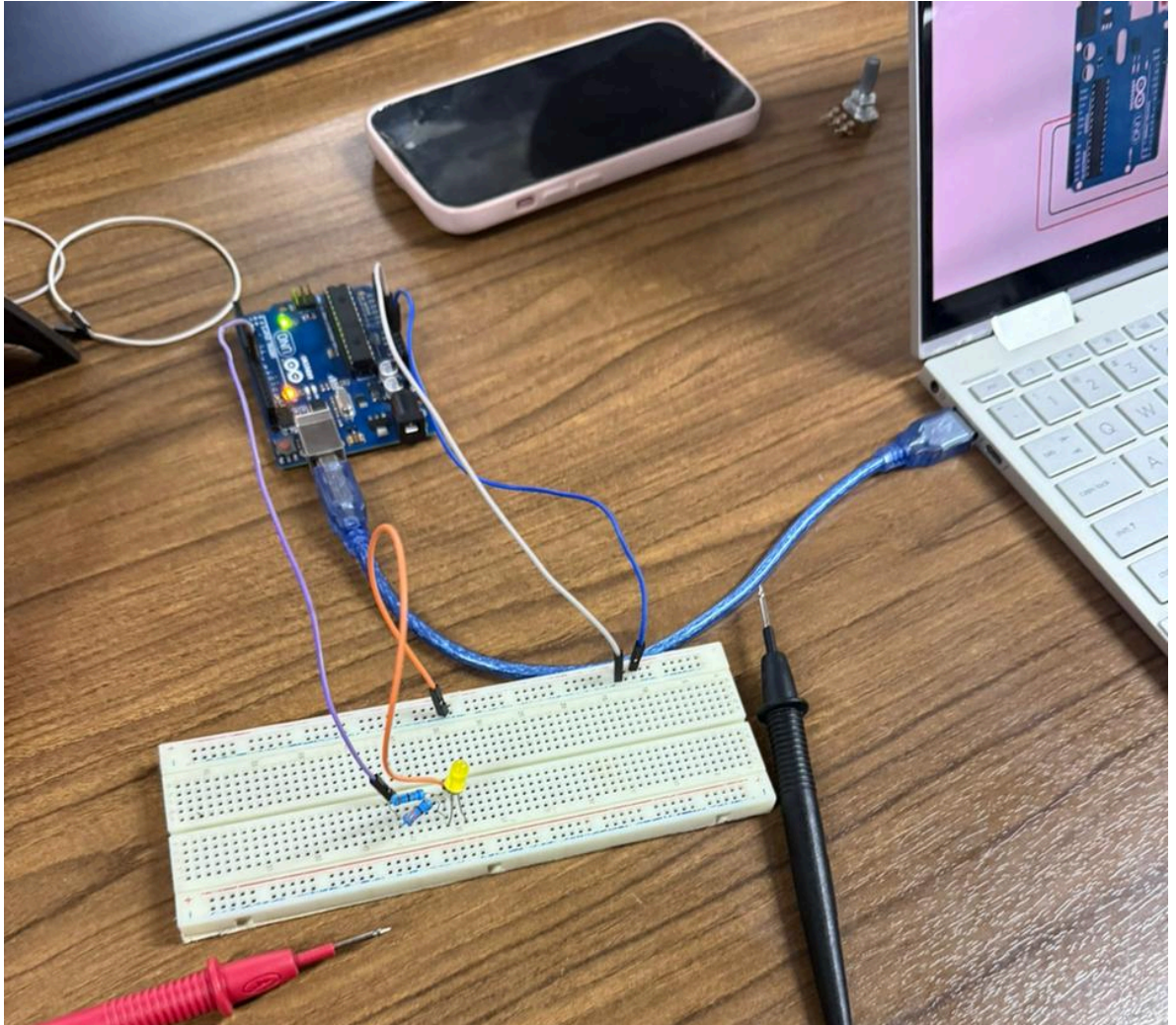
$$\frac{4.751}{650} = \frac{I \times 650}{650}$$

$$I = \underline{\underline{0.0073A}}$$



## Experiment 4

### 1. Physical implementation:

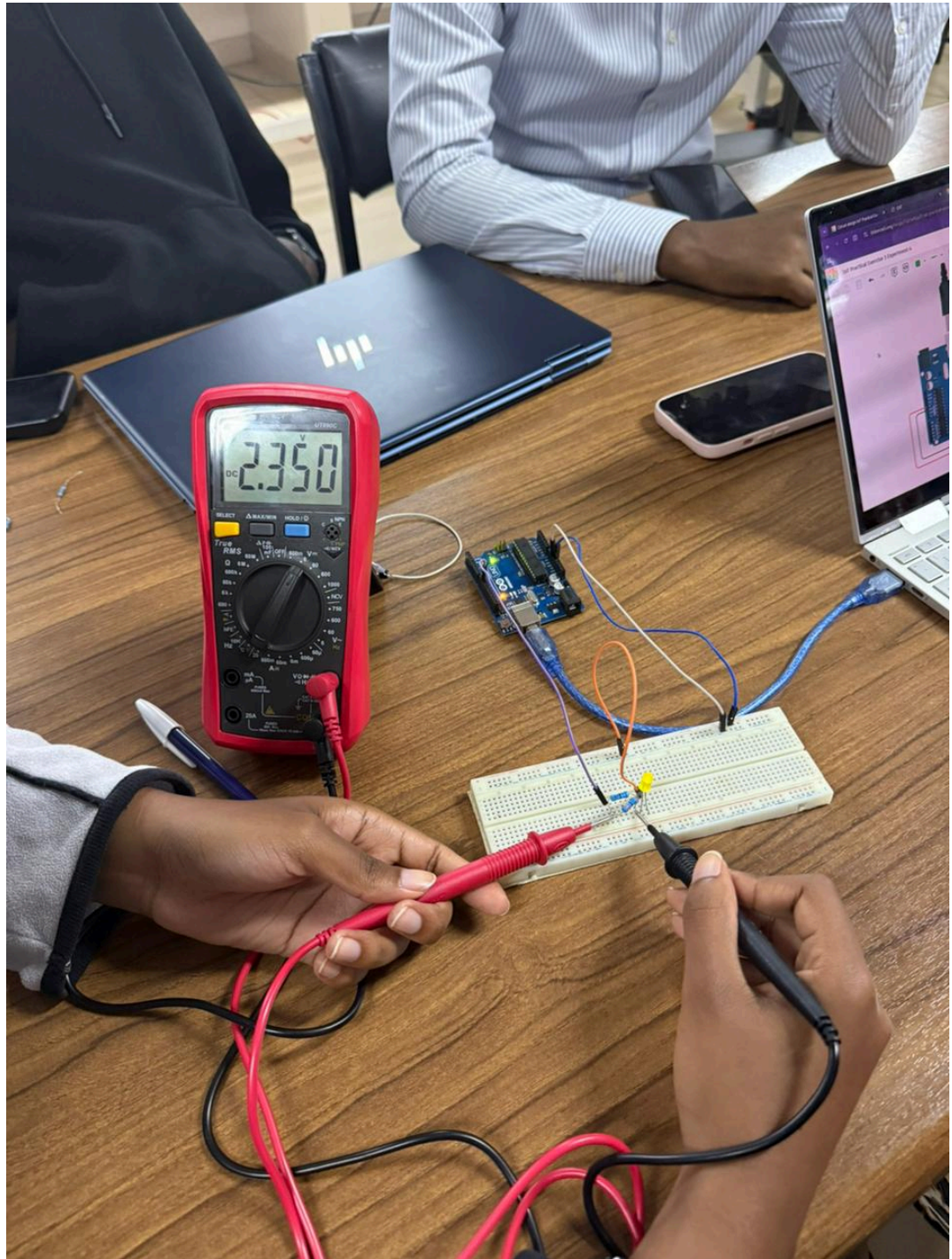


### 2. Blink program deployed:

```
void setup()
{
  pinMode(LED_BUILTIN, OUTPUT);
}

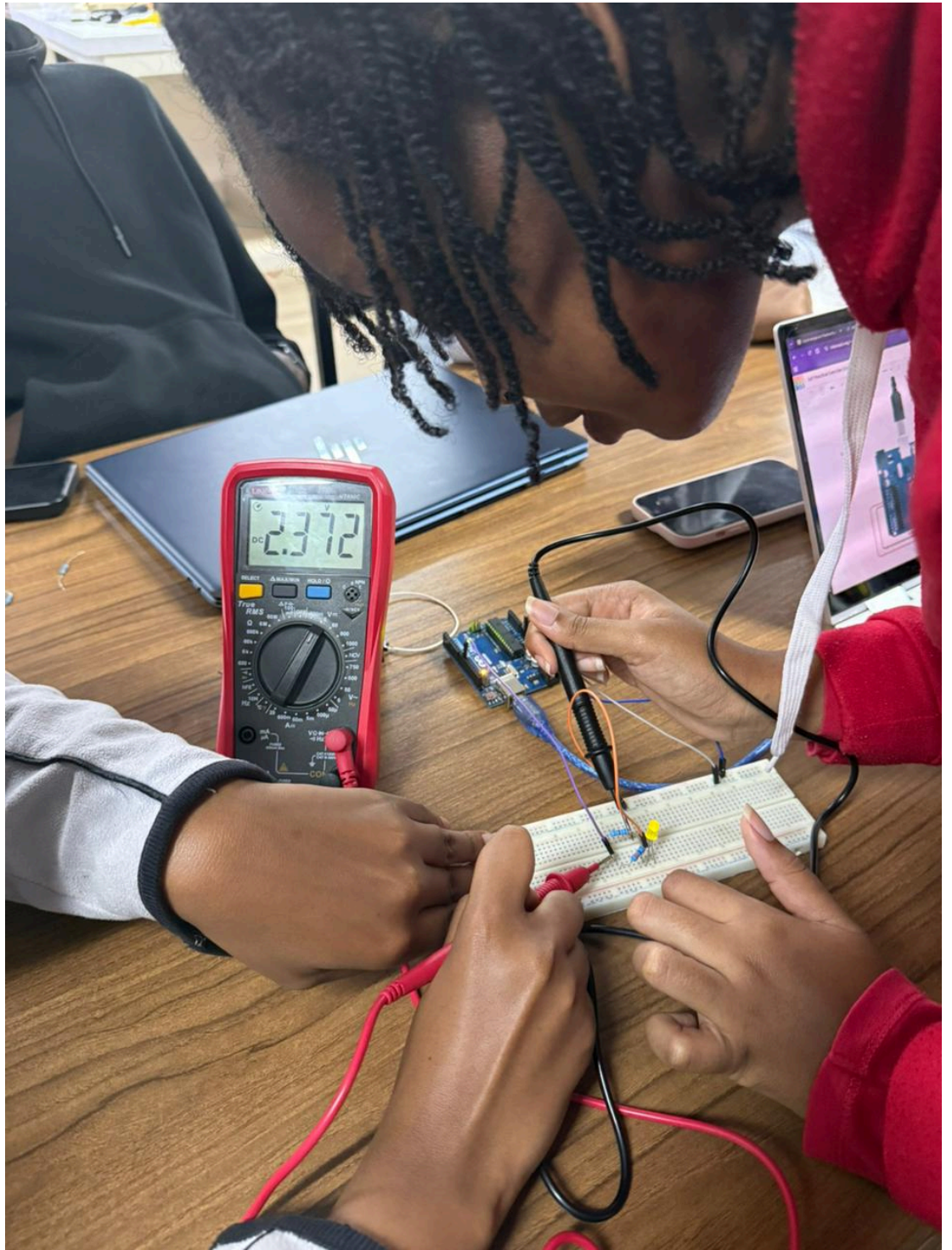
void loop()
{
  digitalWrite(LED_BUILTIN, HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(LED_BUILTIN, LOW);
}
```

3. Voltage at each resistor
  - a. First resistor:





b. Second resistor:



4. Calculations using Ohm's Law:

Experiment 4

i)  $V = IR$

$$\frac{2.350}{330} = \frac{I \times 330}{330}$$

$$I = \underline{\underline{0.0071A}}$$

ii)  $V = IR$

$$\frac{2.372}{650} = \frac{I \times 650}{650}$$

$$I = \underline{\underline{0.0036A}}$$